

Ellipse

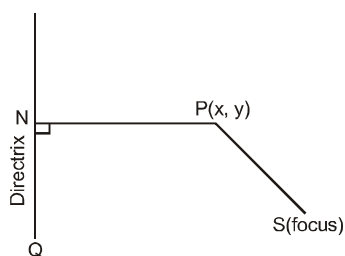
Definitions :

It is locus of a point which moves in such a way that the ratio of its distance from a fixed point and a fixed line (not passes through fixed point and all points and line lies in same plane) is constant (e), which is less than one.

The fixed point is called - **focus**

The fixed line is called - **directrix**.

The constant ratio is called - **eccentricity**, it is denoted by ' e '.



Let S be the focus, QN be the directrix and P be any point on the ellipse. Then, by definition,

$$\frac{PS}{PN} = e \text{ or}$$

$PS = e PN$, $e < 1$, where PN is the length of the perpendicular from P on the directrix QN.

An Alternate Definition An ellipse is the locus of a point that moves in such a way that the sum of its distances from two fixed points (called foci) is constant.

Solved Examples

Ex.1 Find the equation to the ellipse whose focus is the point $(-1, 1)$, whose directrix is the straight line $x - y + 3 = 0$ and eccentricity is $\frac{1}{2}$.

Sol. Let $P \equiv (h, k)$ be moving point,

$$e = \frac{PS}{PM} = \frac{1}{2}$$

$$\Rightarrow (h + 1)^2 + (k - 1)^2$$

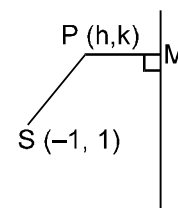
$$= \frac{1}{4} \left(\frac{h - k + 3}{\sqrt{2}} \right)^2$$

$$\Rightarrow \text{locus of } P(h, k) \text{ is}$$

$$8 \{x^2 + y^2 + 2x - 2y + 2\}$$

$$= (x^2 + y^2 - 2xy + 6x - 6y + 9)$$

$$7x^2 + 7y^2 + 2xy + 10x - 10y + 7 = 0.$$



Note :

The general equation of a conic with focus (p, q) & directrix $\ell x + my + n = 0$ is:

$$(\ell^2 + m^2) [(x - p)^2 + (y - q)^2] = e^2 (\ell x + my + n)^2$$

$$\equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

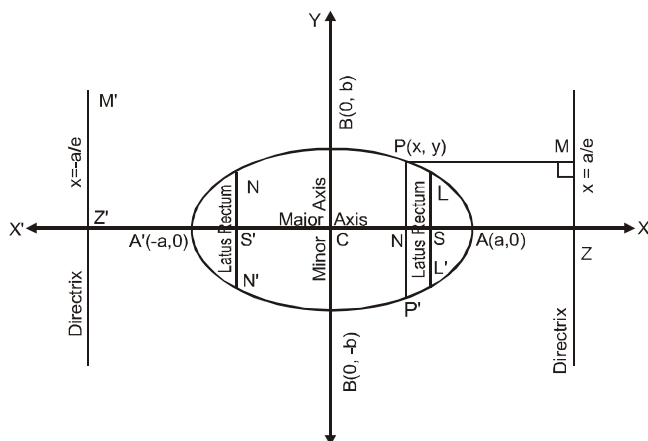
represent ellipse if $0 < e < 1$; $\Delta \neq 0$, $h^2 < ab$

EQUATION OF AN ELLIPSE IN STANDARD FORM

The **Standard form** of the equation of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b),$$

where a and b are constants.



TERMS RELATED TO AN ELLIPSE

A sketch of the locus of a moving point satisfying

the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$, has been shown in the figure given above.

Symmetry

- On replacing y by $-y$, the above equation remains unchanged. So, the curve is symmetrical about x -axis.
- On replacing x by $-x$, the above equation remains unchanged. So, the curve is symmetrical about y -axis

Eccentricity: $e = \sqrt{1 - \frac{b^2}{a^2}}$, $(0 < e < 1)$

Foci: $S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.

Equations of Directrices: $x = \frac{a}{e}$ & $x = -\frac{a}{e}$.

Major Axis: The line segment $A'A$ in which the foci S' & S lie is of length $2a$ & is called the major axis ($a > b$) of the ellipse. Point of intersection of major axis with directrix is called the foot of the directrix (Z).

Minor Axis: The y -axis intersects the ellipse in the points $B' \equiv (0, -b)$ & $B \equiv (0, b)$. The line segment $B'B$ is of length $2b$ ($b < a$) is called the minor axis of the ellipse.

Principal Axis: The major & minor axes together are called principal axis of the ellipse.

Vertices: Point of intersection of ellipse with major axis. $A' \equiv (-a, 0)$ & $A \equiv (a, 0)$.

Focal Chord: A chord which passes through a focus is called a focal chord.

Double Ordinate: A chord perpendicular to the major axis is called a double ordinate.

Latus Rectum: The focal chord perpendicular to the major axis is called the latus rectum.

Length of latus rectum (**LL**)

$$= \frac{2b^2}{a} = \frac{(\text{minor axis})^2}{\text{major axis}} = 2a(1 - e^2)$$

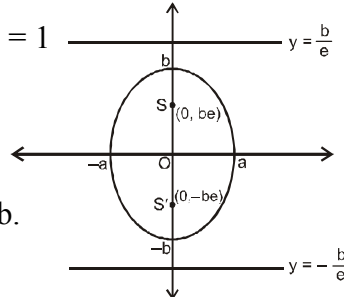
$= 2e$ (distance from focus to the corresponding directrix)

Centre: The point which bisects every chord of the conic drawn through it, is called the centre of the conic. $C \equiv (0, 0)$ the origin is the centre of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Note :

- If the equation of the ellipse is given as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and nothing is mentioned then the rule is to assume that $a > b$.
- If $b > a$ is given, then the y -axis will become major axis and x -axis will become the minor axis and all other points and lines will change accordingly.

Equation: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 

Foci $(0, \pm be)$

Directrices: $y = \pm \frac{b}{e}$

$$a^2 = b^2(1 - e^2), a < b.$$

$$\Rightarrow e = \sqrt{1 - \frac{a^2}{b^2}}$$

Vertices $(0, \pm b)$;

$$l \text{ (L.R.)} = \frac{2a^2}{b},$$

L.R. $y = \pm be$

centre: $(0, 0)$

TWO STANDARD FORMS OF THE ELLIPSE

Standard equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ (a > b)$ (Horizontal Form of an Ellipse)	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \ (a > b)$ (Vertical Form of an ellipse)
Shape of the ellipse		
Centre	(0, 0)	(0, 0)
Equation of major axis	$y = 0$	$x = 0$
Equation of minor axis	$x = 0$	$y = 0$
Length of major axis	$2a$	$2a$
Length of minor axis	$2b$	$2b$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Vertices	$(\pm a, 0)$	$(0, \pm a)$
Equation of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Eccentricity	$e = \sqrt{\frac{a^2 - b^2}{a^2}}$	$e = \sqrt{\frac{a^2 - b^2}{a^2}}$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Ends of latus rectum	$\left(\pm ae, \pm \frac{b^2}{a} \right)$	$\left(\pm \frac{b^2}{a}, \pm ae \right)$
Parametric coordinates	$(a \cos \theta, b \sin \theta)$	$(a \cos \theta, b \sin \theta)$
Focal radius	$P = a - ex_1$ and $S'P = a + ex_1$	$SP = a - ey_1$ and $S'P = a + ey_1$
Sum of focal radii $SP + S'P =$	$2a$	$2a$
Distance between foci	$2ae$	$2ae$
Distance b/w directrices	$\frac{2a}{e}$	$\frac{2a}{e}$
Tangents at the vertices	$x = \pm a$	$y = \pm a$

Solved Examples

Ex.2 Find the centre, the length of the axes, eccentricity and the foci of the ellipse.

$$12x^2 + 4y^2 + 24x - 16y + 25 = 0$$

Sol. The given equation can be written in the form

$$12(x+1)^2 + 4(y-2)^2 = 3 \quad \text{or}$$

$$\frac{(x+1)^2}{1/4} + \frac{(y-2)^2}{3/4} = 1 \quad \dots(1)$$

Co-ordinates of centre of the ellipse are given by

$$x+1=0 \quad \text{and} \quad y-2=0$$

hence centre of the ellipse is $(-1, 2)$

If a and b be the lengths of the semi major and semi minor axes, then $a^2 = 3/4$, $b^2 = 1/4$

$$\therefore \text{Length of major axis} = 2a = \sqrt{3},$$

$$\text{Length of minor axis} = 2b = 1 \quad \therefore a = \frac{\sqrt{3}}{2}, b = \frac{1}{2}$$

$$\text{Since } b^2 = a^2(1 - e^2) \quad \therefore 1/4 = 3/4(1 - e^2)$$

$$\Rightarrow e = \sqrt{2/3} \quad \therefore ae = \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{\sqrt{3}} = \frac{1}{\sqrt{2}},$$

Co-ordinates of foci are given by

$$x+1=0, y-2 = \pm ae$$

$$\text{Thus foci are } \left(-1, 2 \pm \frac{1}{\sqrt{2}}\right)$$

Ex.3 Find the equation of the ellipse having its centre at the point $(2, -3)$ one focus at $(3, -3)$ and one vertex at $(4, -3)$.

Sol. $C \equiv (2, -3)$, $S \equiv (3, -3)$ and $A \equiv (4, -3)$

$$\text{Now } CA = \sqrt{(4-2)^2 + (-3+3)^2} = 2$$

$$\therefore a = 2$$

$$\text{Again } CS = \sqrt{(3-2)^2 + (-3+3)^2} = 1$$

$$\therefore ae = 1 \quad \therefore e = \frac{1}{a} = \frac{1}{2}$$

Let the directrix cut the major-axis at Q .

$$\text{Then } \frac{AS}{AQ} = e = \frac{1}{2}$$

$$\text{If } Q \equiv (\alpha, \beta), \text{ then } SA : AQ = e : 1 = 1 : 2$$

$$\therefore A = \left(\frac{\alpha+6}{3}, \frac{\beta-6}{3}\right) = (4, -3)$$

$$\Rightarrow \frac{\alpha+6}{3} = 4, \quad \frac{\beta-6}{3} = -3 \quad \therefore \alpha = 6, \beta = -3$$

Slope of $CA = 0$, therefore directrix will be parallel to y -axis.

Since directrix is parallel to y -axis and it passes through point $Q(6, -3)$

$$\therefore \text{equation of directrix will be } x = 6$$

Let $P(x, y)$ is a point on ellipse. then

$$e = \frac{1}{2} = \frac{PS}{PN} = \frac{\sqrt{(x-3)^2 + (y+3)^2}}{\left|x - 6/\sqrt{1^2}\right|}$$

$$\therefore \frac{1}{4} = \frac{(x-3)^2 + (y+3)^2}{(x-6)^2}; \quad \text{or}$$

$$(x-6)^2 = 4[(x-3)^2 + (y+3)^2]$$

$$\text{or } x^2 - 12x + 36 = 4[x^2 - 6x + 9 + y^2 + 6y + 9]$$

$$\text{or } 3x^2 + 4y^2 - 12x + 24y + 36 = 0$$

Ex.4 Find the equation to the ellipse whose centre is origin, axes are the axes of co-ordinate and passes through the points $(2, 2)$ and $(3, 1)$.

Sol. Let the equation to the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Since it passes through the points $(2, 2)$ and $(3, 1)$

$$\therefore \frac{4}{a^2} + \frac{4}{b^2} = 1 \quad \dots\dots\dots(i)$$

$$\text{and } \frac{9}{a^2} + \frac{1}{b^2} = 1 \quad \dots\dots\dots(ii)$$

from (i) - 4(ii), we get

$$\frac{4-36}{a^2} = 1-4 \quad \Rightarrow \quad a^2 = \frac{32}{3}$$

from (i), we get

$$\frac{1}{b^2} = \frac{1}{4} - \frac{3}{32} = \frac{8-3}{32} \quad b^2 = \frac{32}{5}$$

$$\therefore \text{Ellipse is } 3x^2 + 5y^2 = 32$$

Ex.5 Find the equation of the ellipse whose foci are (4, 0) and (-4, 0) and eccentricity is $\frac{1}{3}$

Sol. Since both focus lies on x-axis, therefore x-axis is major axis and mid point of foci is origin which is centre and a line perpendicular to major axis and passes through centre is minor axis which is y-axis.

$$\text{Let equation of ellipse is } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore ae = 4 \quad \text{and} \quad e = \frac{1}{3} \text{ (Given)}$$

$$\therefore a = 12 \quad \text{and} \quad b^2 = a^2 (1 - e^2)$$

$$\Rightarrow b^2 = 144 \left(1 - \frac{1}{9}\right)$$

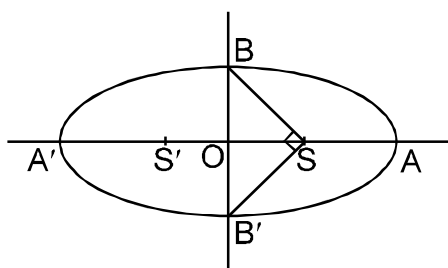
$$b^2 = 16 \times 8$$

$$b = 8\sqrt{2}$$

$$\text{Equation of ellipse is } \frac{x^2}{144} + \frac{y^2}{128} = 1$$

Ex.6 If minor-axis of ellipse subtend a right angle at its focus then find the eccentricity of ellipse.

Sol. Let the equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
($a > b$)



$$\therefore \angle BSB' = \frac{\pi}{2}$$

$$\text{and} \quad OB = OB'$$

$$\therefore \angle BSO = \frac{\pi}{4}$$

$$\Rightarrow OS = OB \quad \Rightarrow \quad ae = b$$

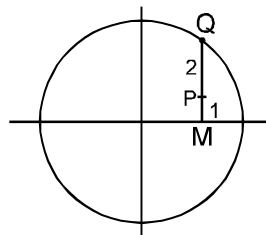
$$\Rightarrow e^2 = \frac{b^2}{a^2} = 1 - e^2 \quad \Rightarrow \quad e = \frac{1}{\sqrt{2}}$$

Ex.7 From a point Q on the circle $x^2 + y^2 = a^2$, perpendicular QM are drawn to x-axis, find the locus of point 'P' dividing QM in ratio 2 : 1.

Sol. Let $Q \equiv (a \cos \theta, a \sin \theta)$

$$M \equiv (a \cos \theta, 0)$$

Let $P \equiv (h, k)$



$$\therefore h = a \cos \theta, k = \frac{a \sin \theta}{3}$$

$$\therefore \left(\frac{3k}{a}\right)^2 + \left(\frac{h}{a}\right)^2 = 1$$

$$\Rightarrow \text{Locus of P is } \frac{x^2}{a^2} + \frac{y^2}{(a/3)^2} = 1$$

Ex.8 Find the equation of axes, directrix, co-ordinate of foci, centre, vertices, length of latus-rectum and eccentricity of an ellipse $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{16} = 1$.

Sol. Let $x - 3 = X$, $y - 2 = Y$, so equation of ellipse

$$\text{becomes as } \frac{X^2}{5^2} + \frac{Y^2}{4^2} = 1.$$

$$\text{equation of major axis is } Y = 0 \quad \Rightarrow y = 2.$$

$$\text{equation of minor axis is } X = 0 \quad \Rightarrow x = 3.$$

$$\text{centre } (X = 0, Y = 0) \quad \Rightarrow x = 3, y = 2$$

$$C \equiv (3, 2)$$

$$\text{Length of semi-major axis } a = 5$$

$$\text{Length of major axis } 2a = 10$$

$$\text{Length of semi-minor axis } b = 4$$

$$\text{Length of minor axis } = 2b = 8.$$

Let 'e' be eccentricity

$$\therefore b^2 = a^2 (1 - e^2)$$

$$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{\frac{25 - 16}{25}} = \frac{3}{5}.$$

$$\text{Length of latus rectum} = LL' = \frac{2b^2}{a} = \frac{2 \times 16}{5} = \frac{32}{5}$$

$$\text{Co-ordinates foci are } X = \pm ae, Y = 0$$

$$\Rightarrow S \equiv (X = 3, Y = 0) \quad \& \quad S' \equiv (X = -3, Y = 0)$$

$$\Rightarrow S \equiv (6, 2) \quad \& \quad S' \equiv (0, 2)$$

Co-ordinate of vertices

Extremities of major axis

$$A \equiv (X = a, Y = 0) \quad \& \quad A' \equiv (X = -a, Y = 0)$$

$$\Rightarrow A \equiv (x = 8, y = 2) \quad \& \quad A' \equiv (x = -2, y = 2)$$

$$A \equiv (8, 2) \quad \& \quad A' \equiv (-2, 2)$$

Extremities of minor axis

$$B \equiv (X = 0, Y = b) \quad \& \quad B' \equiv (X = 0, Y = -b)$$

$$B \equiv (x = 3, y = 6) \quad \& \quad B' \equiv (x = 3, y = -2)$$

$$B \equiv (3, 6) \quad \& \quad B' \equiv (3, -2)$$

Equation of directrix $X = \pm \frac{a}{e}$

$$x - 3 = \pm \frac{25}{3} \quad \Rightarrow \quad x = \frac{34}{3} \quad \& \quad x = -\frac{16}{3}$$

Auxiliary Circle

The circle drawn on major axis AA' as diameter is known as the Auxiliary circle.

Let the equation of

the ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Then the equation of

its auxiliary circle is $x^2 + y^2 = a^2$.

Let Q be a point on auxiliary circle so that QM , perpendicular to major axis meets the ellipse at P . The points P and Q are called as corresponding point on the ellipse and auxiliary circle respectively.

The angle θ is known as eccentric angle of the point P on the ellipse.

It may be noted that the CQ and not CP is inclined at θ with x -axis.

Note that :

$$\frac{\ell(PM)}{\ell(QM)} = \frac{b}{a} = \frac{\text{Semi minor axis}}{\text{Semi major axis}}$$

NOTE :

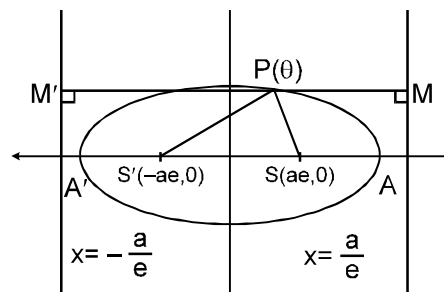
If from each point of a circle perpendiculars are drawn upon a fixed diameter then the locus of the points dividing these perpendiculars in a given ratio is an ellipse of which the given circle is the auxiliary circle.

Solved Examples

Ex.9 Find the focal distance of a point $P(\theta)$ on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$$

Sol. Let 'e' be the eccentricity of ellipse.



$$\therefore PS = e \cdot PM = e \left(\frac{a}{e} - a \cos \theta \right)$$

$$PS = (a - ae \cos \theta)$$

and $PS' = e \cdot PM' = e \left(a \cos \theta + \frac{a}{e} \right)$

$$PS' = a + ae \cos \theta$$

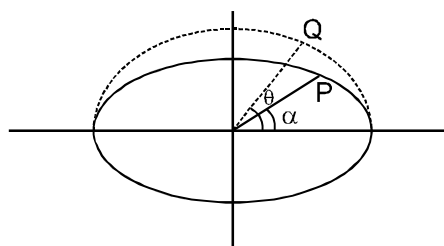
$$\therefore \text{focal distance are } (a \pm ae \cos \theta)$$

Note : $PS + PS' = 2a$ $PS + PS' = AA'$

Ex.10 Find the distance from centre of the point P on the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose radius makes angle α with x -axis.

Sol. Let $P \equiv (a \cos \theta, b \sin \theta)$



$$\therefore m_{(op)} = \frac{b}{a} \tan \theta = \tan \alpha \Rightarrow \tan \theta = \frac{a}{b} \tan \alpha$$

$$OP = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = \sqrt{\frac{a^2 + b^2 \tan^2 \theta}{\sec^2 \theta}}$$

$$= \sqrt{\frac{a^2 + b^2 \tan^2 \theta}{1 + \tan^2 \theta}} = \sqrt{\frac{a^2 + b^2 \times \frac{a^2}{b^2} \tan^2 \alpha}{1 + \frac{a^2}{b^2} \tan^2 \alpha}}$$

$$OP = \frac{ab}{\sqrt{a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}}$$

PARAMETRIC EQUATION OF THE ELLIPSE

The coordinates $x = a \cos \theta$ and $y = b \sin \theta$ satisfy the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

for all real values of θ . Thus, $x = a \cos \theta$, $y = b \sin \theta$ are the parametric equation of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ where the parameter } 0 \leq \theta < 2\pi.$$

Hence the coordinates of any point on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

may be taken as $(a \cos \theta, b \sin \theta)$. This point is also called the point ' θ '.

The angle θ is called the eccentric angle of the point $(a \cos \theta, b \sin \theta)$ on the ellipse.

Equation of Chord

The equation of the chord joining the points $P \equiv (a \cos \theta_1, b \sin \theta_1)$ and $Q \equiv (a \cos \theta_2, b \sin \theta_2)$ is

$$\frac{x}{a} \cos \left(\frac{\theta_1 + \theta_2}{2} \right) + \frac{y}{b} \sin \left(\frac{\theta_1 + \theta_2}{2} \right) = \cos \left(\frac{\theta_1 - \theta_2}{2} \right).$$

Solved Examples

Ex.11 Write the equation of chord of an ellipse

$$\frac{x^2}{25} + \frac{y^2}{16} = 1 \text{ joining two points } P \left(\frac{\pi}{4} \right) \text{ and } Q \left(\frac{5\pi}{4} \right).$$

Sol. Equation of chord is $\frac{x}{5} \cos \left(\frac{\frac{\pi}{4} + \frac{5\pi}{4}}{2} \right) + \frac{y}{4} \sin \left(\frac{\frac{\pi}{4} + \frac{5\pi}{4}}{2} \right) = \cos \left(\frac{\frac{\pi}{4} - \frac{5\pi}{4}}{2} \right)$

$$\sin \frac{\left(\frac{\pi}{4} + \frac{5\pi}{4} \right)}{2} = \cos \frac{\left(\frac{\pi}{4} - \frac{5\pi}{4} \right)}{2}$$

$$\frac{x}{5} \cdot \cos \left(\frac{3\pi}{4} \right) + \frac{y}{4} \cdot \sin \left(\frac{3\pi}{4} \right) = 0$$

$$-\frac{x}{5} + \frac{y}{4} = 0 \quad \Rightarrow \quad 4x = 5y$$

Ex.12 If $P(\alpha)$ and $P(\beta)$ are extremities of a focal chord of ellipse then prove that its eccentricity

$$e = \left| \frac{\cos \left(\frac{\alpha - \beta}{2} \right)}{\cos \left(\frac{\alpha + \beta}{2} \right)} \right|.$$

Sol. Let the equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore \text{ equation of chord is } \frac{x}{a} \cos \left(\frac{\alpha + \beta}{2} \right) + \frac{y}{b} \sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

Since above chord is focal chord,

$\therefore$ it passes through focus $(ae, 0)$ or $(-ae, 0)$

$$\therefore \pm e \cos \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\therefore e = \left| \frac{\cos \left(\frac{\alpha - \beta}{2} \right)}{\cos \left(\frac{\alpha + \beta}{2} \right)} \right| \quad \text{Ans.}$$

Note :

$$\therefore \pm e = \frac{\cos \frac{\alpha - \beta}{2}}{\cos \frac{\alpha + \beta}{2}} \Rightarrow \pm e = \frac{1 + \tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}}$$

Applying componendo and dividendo

$$\frac{1 \pm e}{\pm e - 1} = \frac{2}{2 \tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}}$$

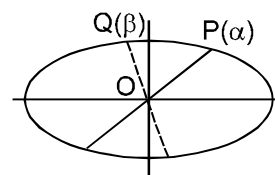
$$\tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{1+e}{e-1} \text{ or } \frac{e-1}{1+e}$$

Ex.13 Find the angle between two diameters of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \text{ Whose extremities have eccentricity}$$

$$\text{angle } \alpha \text{ and } \beta = \alpha + \frac{\pi}{2}.$$

Sol. Let ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



$$\text{Slope of OP} = m_1 = \frac{b \sin \alpha}{a \cos \alpha} = \frac{b}{a} \tan \alpha$$

$$\text{Slope of OQ} = m_2 = \frac{b \sin \beta}{a \cos \beta} = -\frac{b}{a} \cot \alpha$$

$$\text{given } \beta = \alpha + \frac{\pi}{2}$$

$$\begin{aligned} \therefore \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{b}{a}(\tan \alpha + \cot \alpha)}{1 - \frac{b^2}{a^2}} \right| \\ &= \left| \frac{2ab}{(a^2 - b^2) \sin 2\alpha} \right| \end{aligned}$$

POSITION OF A POINT WITH RESPECT TO AN ELLIPSE

The point $P(x_1, y_1)$ lies outside, on or inside the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ according as}$$

$$\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 > 0, = 0 \text{ or } < 0.$$

Solved Examples

Ex.14 Check whether the point $P(3, 2)$ lies inside or outside of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

$$\text{Sol. } S_1 = \frac{9}{25} + \frac{4}{16} - 1 = \frac{9}{25} + \frac{1}{4} - 1 < 0$$

$\therefore$ Point $P \equiv (3, 2)$ lies inside the ellipse.

Ex.15 Find the set of value(s) of ' α ' for which the point

$$P(\alpha, -\alpha) \text{ lies inside the ellipse } \frac{x^2}{16} + \frac{y^2}{9} = 1.$$

Sol. If $P(\alpha, -\alpha)$ lies inside the ellipse

$$\therefore S_1 < 0$$

$$\Rightarrow \frac{\alpha^2}{16} + \frac{\alpha^2}{9} - 1 < 0$$

$$\Rightarrow \frac{25}{144} \cdot \alpha^2 < 1 \quad \Rightarrow \quad \alpha^2 < \frac{144}{25}$$

$$\therefore \alpha \in \left(-\frac{12}{5}, \frac{12}{5} \right).$$

CONDITION OF TANGENCY AND POINT OF CONTACT

The condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is that $c^2 = a^2 m^2 + b^2$ and the coordinates of the points of contact are

$$\left(\pm \frac{a^2 m}{\sqrt{a^2 m^2 + b^2}}, \mp \frac{b^2}{\sqrt{a^2 m^2 + b^2}} \right)$$

Note

* $x \cos a + y \sin a = p$ is a tangent if

$$p^2 = a^2 \cos^2 \alpha + b^2 \sin^2 \alpha.$$

* $lx + my + n = 0$ is a tangent if $n^2 = a^2 l^2 + b^2 m^2$.

Solved Examples

Ex.16 Find the set of value(s) of ' λ ' for which the line

$$3x - 4y + \lambda = 0 \text{ intersect the ellipse}$$

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \text{ at two distinct points.}$$

Sol. Solving given line with ellipse,

$$\text{we get } \frac{(4y - \lambda)^2}{9 \times 16} + \frac{y^2}{9} = 1$$

$$\frac{2y^2}{9} - \frac{y\lambda}{18} + \frac{\lambda^2}{144} - 1 = 0$$

Since, line intersect the parabola at two distinct points,

$\therefore$ roots of above equation are real & distinct

$$\therefore D > 0$$

$$\Rightarrow \frac{\lambda^2}{(18)^2} - \frac{8}{9} \cdot \left(\frac{\lambda^2}{144} - 1 \right) > 0$$

$$\Rightarrow -12\sqrt{2} < \lambda < 12\sqrt{2}$$

Tangents :

- (a) Slope form: $y = mx \pm \sqrt{a^2 m^2 + b^2}$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for all values of m .
- (b) Point form: $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1) .
- (c) Parametric form: $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(a \cos \theta, b \sin \theta)$.

Note :

- (i) There are two tangents to the ellipse having the same m , i.e. there are two tangents parallel to any given direction. These tangents touch the ellipse at extremities of a diameter.
- (ii) Point of intersection of the tangents at the point α & β is, $\left(a \frac{\cos \frac{\alpha+\beta}{2}}{\cos \frac{\alpha-\beta}{2}}, b \frac{\sin \frac{\alpha+\beta}{2}}{\cos \frac{\alpha-\beta}{2}} \right)$
- (iii) The eccentric angles of the points of contact of two parallel tangents differ by π .

Solved Examples

Ex.17 Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

Sol. Slope of tangent $= m = \frac{1}{2}$

Given ellipse is $\frac{x^2}{4} + \frac{y^2}{3} = 1$

Equation of tangent whose slope is ' m ' is

$$y = mx \pm \sqrt{4m^2 + 3}$$

$$\therefore m = \frac{1}{2}$$

$$\therefore y = \frac{1}{2}x \pm \sqrt{1+3}$$

$$2y = x \pm 4$$

Ex.18 For what value of λ does the line $y = x + \lambda$ touch the ellipse $9x^2 + 16y^2 = 144$.

Sol. $\therefore$ Equation of ellipse is

$$9x^2 + 16y^2 = 144 \quad \text{or} \quad \frac{x^2}{16} + \frac{y^2}{9} = 1$$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

then we get $a^2 = 16$ and $b^2 = 9$

and comparing the line $y = x + \lambda$ with $y = mx + c$
 $\therefore m = 1$ and $c = \lambda$

If the line $y = x + \lambda$ touches the ellipse

$$9x^2 + 16y^2 = 144, \text{ then } c^2 = a^2 m^2 + b^2$$

$$\Rightarrow \lambda^2 = 16 \times 1^2 + 9 \Rightarrow \lambda^2 = 25 \quad \therefore \lambda = \pm 5$$

Ex.19 Find the equation of tangent to the ellipse $4x^2 + 9y^2 = 36$ at the point $(3, -2)$.

Sol. We have $4x^2 + 9y^2 = 36 \Rightarrow \frac{x^2}{9} + \frac{y^2}{4} = 1$.

This is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

where $a^2 = 9$ and $b^2 = 4$.

We know that the equation of the tangent to the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

So, the equation of the tangent to the given ellipse at $(3, -2)$ is

$$\frac{3x}{9} - \frac{2y}{4} = 1 \quad \text{i.e.,} \quad \frac{x}{3} - \frac{y}{2} = 1$$

Ex.20 Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

Sol. Let m be the slope of the tangent, since the tangent is perpendicular to the line $y + 2x = 4$.

$$\therefore mx - 2 = -1 \Rightarrow m = \frac{1}{2}$$

$$\text{Since } 3x^2 + 4y^2 = 12 \quad \text{or} \quad \frac{x^2}{4} + \frac{y^2}{3} = 1$$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore a^2 = 4$ and $b^2 = 3$

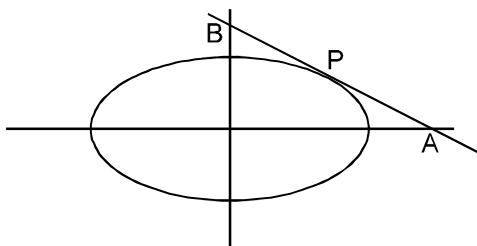
So the equation of the tangent are

$$y = \frac{1}{2}x \pm \sqrt{4 \times \frac{1}{2} + 3}$$

$$\Rightarrow y = \frac{1}{2}x \pm 2 \quad \text{or} \quad x - 2y \pm 4 = 0$$

Ex.21 A tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ touches at the point P on it in the first quadrant and meets the co-ordinate axes in A and B respectively. If P divides AB in the ratio 3 : 1, find the equation of the tangent.

Sol. Let $P \equiv (a \cos \theta, b \sin \theta)$



$\therefore$ equation of tangent is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$A \equiv (a \sec \theta, 0) \quad B \equiv (0, b \operatorname{cosec} \theta)$$

$\therefore$ P divide AB internally in the ratio 3 : 1

$$\therefore a \cos \theta = \frac{a \sec \theta}{4}$$

$$\Rightarrow \cos^2 \theta = \frac{1}{4} \quad \Rightarrow \quad \cos \theta = \frac{1}{2} \text{ and}$$

$$b \sin \theta = \frac{3b \operatorname{cosec} \theta}{4} \quad \Rightarrow \quad \sin \theta = \frac{\sqrt{3}}{2}$$

$$\therefore \text{tangent is } \frac{x}{2a} + \frac{\sqrt{3}y}{2b} = 1$$

$$\Rightarrow bx + \sqrt{3} ay = 2ab$$

Ex.22 Prove that the locus of the point of intersection of tangents to an ellipse at two points whose eccentric angle differ by a constant α is an ellipse.

Sol. Let P (h, k) be the point of intersection of tangents at A(θ) and B(β) to the ellipse.

$$\therefore h = \frac{a \cos \left(\frac{\theta + \beta}{2} \right)}{\cos \left(\frac{\theta - \beta}{2} \right)} \quad \& \quad k = \frac{b \sin \left(\frac{\theta + \beta}{2} \right)}{\cos \left(\frac{\theta - \beta}{2} \right)}$$

$$\Rightarrow \left(\frac{h}{a} \right)^2 + \left(\frac{k}{b} \right)^2 = \sec^2 \left(\frac{\theta - \beta}{2} \right)$$

but given that $\theta - \beta = \alpha$

$$\therefore \text{locus is } \frac{x^2}{a^2 \sec^2 \left(\frac{\alpha}{2} \right)} + \frac{y^2}{b^2 \sec^2 \left(\frac{\alpha}{2} \right)} = 1$$

Ex.23 Find the locus of foot of perpendicular drawn from

centre to any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Sol. Let P(h, k) be the foot of perpendicular to a tangent

$$y = mx + \sqrt{a^2 m^2 + b^2} \quad \text{.....(i)}$$

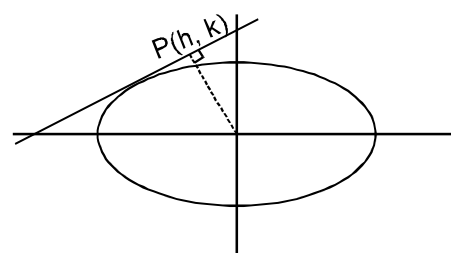
from centre

$$\therefore \frac{k}{h} \cdot m = -1 \quad \Rightarrow \quad m = -\frac{h}{k} \quad \text{.....(ii)}$$

$\therefore$ P(h, k) lies on tangent

$$\therefore k = mh + \sqrt{a^2 m^2 + b^2} \quad \text{.....(iii)}$$

from equation (ii) & (iii), we get



$$\left(k + \frac{h^2}{k} \right)^2 = \frac{a^2 h^2}{k^2} + b^2$$

$$\Rightarrow \text{locus is } (x^2 + y^2)^2 = a^2 x^2 + b^2 y^2$$

EQUATION OF NORMAL IN DIFFERENT FORMS

(i) Point Form

The equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

at the point (x_1, y_1) is $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$.

(ii) Parametric Form

The equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

at the point ($a \cos \theta, b \sin \theta$) is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2.$$

$$\text{or } \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2.$$

(iii) Slope Form

The equation of normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

in terms of slope 'm' is $y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2 m^2}}$